

Impact of Uncertainty on Economic Dispatch under Time-Sequential Operation with Practical Constraints

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Abstract:

Economic Dispatch (ED) plays an important role in short-term power system operation by determining the optimal generation schedule that minimizes operating cost while satisfying system constraints. Traditional ED models are generally formulated as deterministic optimization problems, where forecast demand is assumed to be perfectly known. However, in practical power system operation, real-time demand often deviates from forecast values due to load uncertainty and renewable variability.

This work analyzes the impact of uncertainty on economic dispatch under a time-sequential operational framework consisting of Day-Ahead (DA) scheduling and Real-Time (RT) adjustments. A simplified 2-bus test system is modeled using AMES software to study system behavior under different reserve levels. Day-ahead and real-time load data for three consecutive days are used to simulate uncertainty conditions.

Different reserve levels such as 0%, 10%, 15%, and 50% are analyzed to study system feasibility. Simulation results show that low reserve levels improve system flexibility but fail under high uncertainty conditions. Although 50% reserve ensures full feasibility, it is not practical due to higher operational cost. The study highlights the need for adaptive reserve allocation under uncertain operating conditions.

Introduction:

Economic Dispatch (ED) is a fundamental optimization problem in power system operation that determines the optimal power output of available generating units while minimizing total operating cost. The objective of ED is to ensure that total generation meets system demand while satisfying operational constraints such as generator capacity limits, power balance requirements, ramping constraints, and transmission limits.

Traditionally, ED is solved as a deterministic optimization problem where system demand is assumed to be accurately known in advance.

However, modern power systems are increasingly affected by uncertainty due to renewable energy penetration, load forecast errors, and market variability. These uncertainties create deviations between scheduled generation and actual demand.

In practical systems, operation occurs in a time-sequential manner. Day-ahead scheduling is performed based on forecast demand, while real-time operation requires adjustments based on actual system conditions. If adequate reserve is not available during real-time operation, the system may become in-feasible.

This work focuses on analyzing how uncertainty impacts system operation under time-sequential economic dispatch and how reserve affects system feasibility.

Literature review:

Several research works have studied economic dispatch under uncertain operating conditions using stochastic optimization, robust optimization, and scenario-based methods.

Lamichhane and Dubey (2025) proposed a two-stage stochastic dynamic economic dispatch model with battery energy storage systems under wind and load uncertainty. Their work demonstrated the importance of recourse actions in reducing operating cost.

Gu et al. (2022) developed chance-constrained and robust optimization approaches for economic dispatch under renewable uncertainty. Their study analyzed trade-offs between reliability and operational cost.

Zeng et al. (2020) introduced scenario generation and reduction techniques for economic dispatch involving wind power and energy storage systems. Their work reduced computational complexity while maintaining solution quality.

Chen et al. (2023) proposed robust optimization-based reserve allocation under uncertain load conditions.

Although these works effectively model uncertainty within optimization frameworks, most of them do not explicitly analyze real-time operational behavior under time-sequential scheduling.

Research gap:

Existing research primarily focuses on solving economic dispatch problems using advanced optimization techniques under uncertain conditions. However, most studies formulate uncertainty within a single optimization framework.

Very limited work has been done to analyze actual time-sequential operation involving day-ahead scheduling followed by real-time correction.

Additionally, operational constraints such as reserve constraints are often included mathematically but their impact on real-time feasibility is not studied explicitly.

This work addresses this gap by analyzing system behavior under real-time uncertainty using sequential simulation.

Objectives:

The objectives of this work are:

- To study the impact of uncertainty on economic dispatch operation
- To analyze time-sequential operation involving Day-Ahead and Real-Time scheduling
- To evaluate the role of reserve in handling uncertainty
- To study system feasibility under different reserve levels
- To identify limitations of fixed reserve allocation

Mathematical formulation:

Objective function

$$\min \sum_{i=1}^N C_i(P_{gi})$$

Where C_i = Cost for i^{th} generator

P_{gi} = Power generation for i^{th} generator

subject to:

$$0 \leq P_{gi} \leq P_{gi}^{\max} ; \forall i = 1 \dots N$$

$$-R_{id} \leq P_{gi}^t - P_{gi}^{t-1} \leq R_{iup} ; \forall i = 1 \dots N$$

$$-p_{ab}^{\min} \leq p_{ab} \leq p_{ab}^{\max} ; a \text{ \& \& } b \text{ are buses}$$

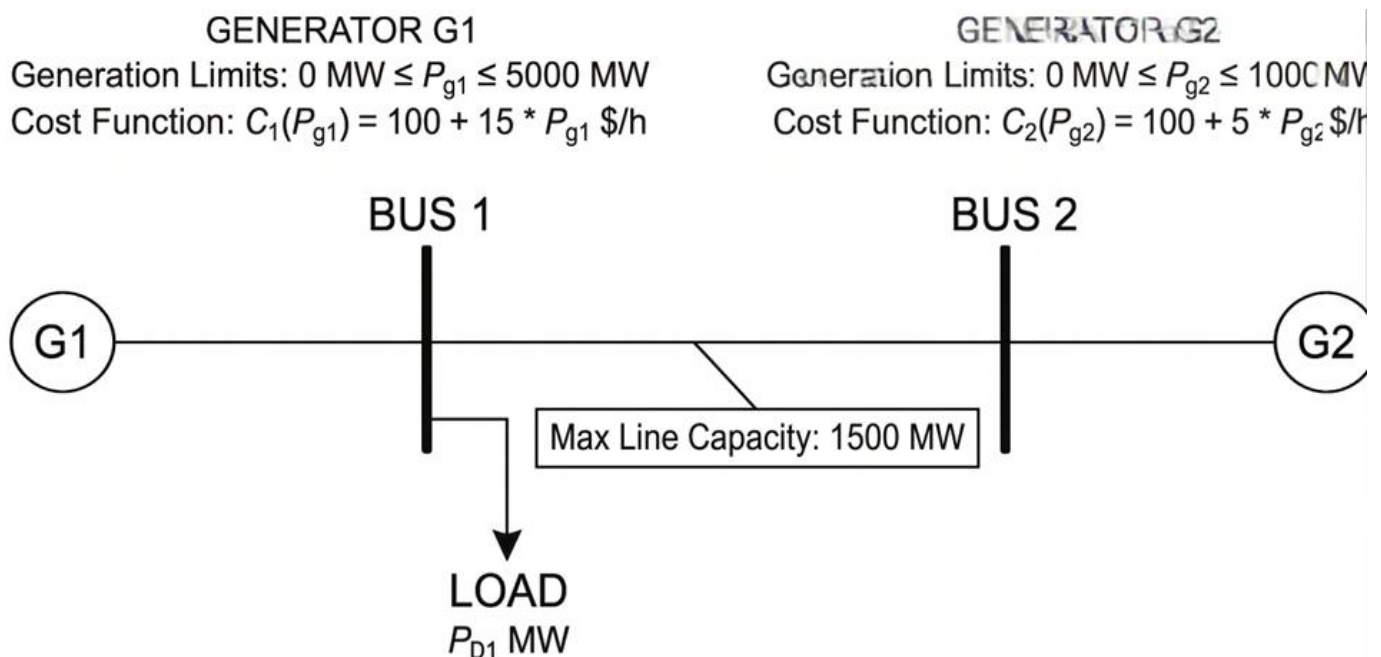
$$\sum_{i=1}^N P_{gi}^{\max} - \sum_{i=1}^N P_{gi} \geq R ; R \text{ is reserve}$$

Where R_{id} , R_{iup} = Ramping constraints for i^{th} generator

P_{ab} = Power flows between buses a and b

System description:

A simplified 2-bus power system is considered in this study for better understanding of uncertainty behavior.



The system consists of:

- Two generating units
- One transmission line
- One load bus

Generator 1 is modeled as a low-cost generator with higher capacity, while Generator 2 is modeled as a relatively expensive generator.

The simplified system helps isolate the effect of uncertainty without introducing unnecessary network complexity.

Data used:

Day-ahead and real-time load data for three consecutive days were used in this study.

The day-ahead load represents forecasted demand values used for scheduling.

Real-time load represents actual demand values used during real-time operation.

To create uncertainty scenarios, load values were scaled to generate variations between day-ahead and real-time demand.

The observed variation ranged from approximately:

Minimum: -23%

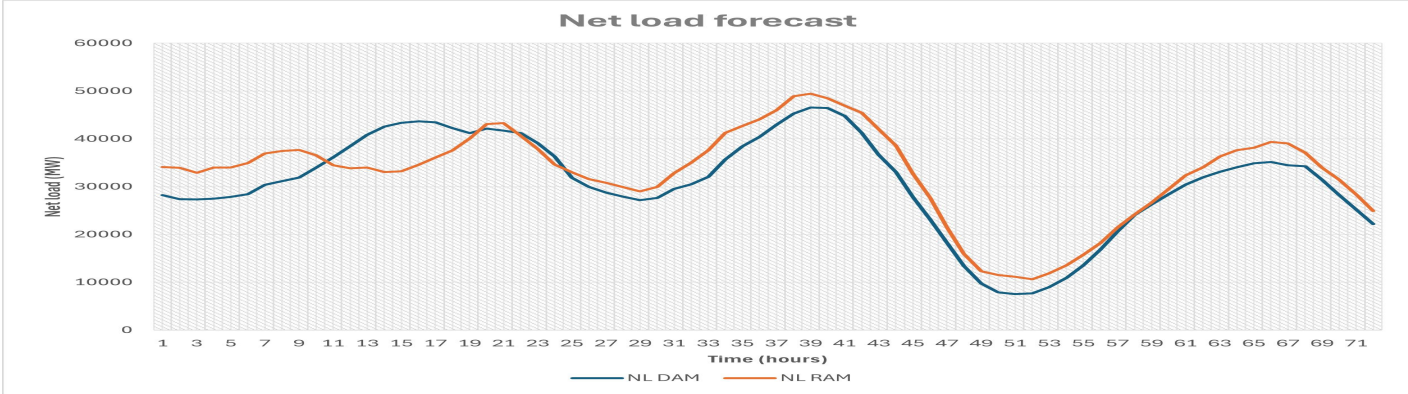
Maximum: +48%

Day0	DAM-Wind	RTM-Wind	% variation		DAM-Load	RTM-Load	% variation	SF	NL DAM	NL RAM	% variation		NL DAM (scaled)	NL RAM(scaled)		10% Reserve	15%Rsrv
1.00	9696	8534	-11.98		47604	51183	7.52	2	28213	34115	20.92		705	853		776	811
2.00	9095	7472	-17.84		45569	48873	7.25	2	27378	33929	23.92		684	848		753	787
3.00	8442	7154	-15.25		44229	47200	6.72	2	27344	32892	20.29		684	822		752	786
4.00	7889	6098	-22.70		43266	46184	6.74	2	27488	33988	23.64		687	850		756	790
5.00	7614	6101	-19.87		43104	46182	7.14	2	27876	33980	21.90		697	849		767	801
6.00	7872	6109	-22.40		44193	47140	6.67	2	28449	34922	22.75		711	873		782	818
7.00	8000	5887	-26.41		46397	48696	4.95	2	30397	36921	21.46		760	923		836	874
8.00	8227	5660	-31.20		47606	48787	2.48	2	31152	37468	20.27		779	937		857	896
9.00	8361	5662	-32.29		48595	49021	0.87	2	31873	37698	18.27		797	942		877	916
10.00	8440	6526	-22.67		50828	49601	-2.41	2	33949	36549	7.66		849	914		934	976
11.00	8508	8037	-5.53		53146	50539	-4.91	2	36131	34465	-4.61		903	862		994	1039
12.00	8480	8905	5.00		55401	51638	-6.79	2	38440	33829	-12.00		961	846		1057	1105
13.00	8285	9374	13.14		57394	52725	-8.13	2	40824	33978	-16.77		1021	849		1123	1174
14.00	8371	10397	24.20		59294	53832	-9.21	2	42553	33038	-22.36		1064	826		1170	1223
15.00	8762	10677	21.86		60866	54567	-10.35	2	43343	33212	-23.37		1084	830		1192	1246
16.00	9112	10038	10.16		61874	54669	-11.65	2	43651	34593	-20.75		1091	865		1200	1255
17.00	9471	9600	1.36		62371	55276	-11.38	2	43429	36077	-16.93		1086	902		1194	1249
18.00	9642	8889	-7.81		61548	55338	-10.09	2	42264	37560	-11.13		1057	939		1162	1215
19.00	9067	7201	-20.58		59339	54476	-8.20	2	41205	40073	-2.75		1030	1002		1133	1185
20.00	8026	5423	-32.44		58182	53922	-7.32	2	42130	43076	2.25		1053	1077		1159	1211
21.00	7389	5144	-30.39		56493	53597	-5.13	2	41714	43310	3.82		1043	1083		1147	1199

22.00	6582	5629	-14.48		54367	51886	-4.56	2	41202	40627	-1.40		1030	1016		1133	1185
23.00	6089	5745	-5.65		51250	49382	-3.65	2	39072	37891	-3.02		977	947		1074	1123
24.00	5746	5877	2.28		47779	46374	-2.94	2	36287	34621	-4.59		907	866		998	1043
max	9696	10677	24.20		62371	55338	7.52		43651	43310	23.92		1091	1083			
min	5746	5144	-32.44		43104	46182	-11.65		27344	32892	-23.37		684	822			

Day 1	DAM-Wind	RTM-Wind	% variation		DAM-Load	RTM-Load	% variation	SF	NL DAM	NL RAM	% variation		NL DAM (scaled)	NL RAM(scaled)		10% Reserve	15%Rsrv
1.00	6006	5457	-9.14		43858	43919	0.14	2	31846	33005	3.64		1062	866		1168	1221
2.00	5884	5224	-11.21		41741	42016	0.66	2	29974	31568	5.32		999	1052		1099	1149
3.00	5729	5017	-12.43		40218	40838	1.54	2	28760	30805	7.11		959	1027		1055	1102
4.00	5705	5098	-10.65		39340	40083	1.89	2	27929	29888	7.01		931	996		1024	1071
5.00	5980	5555	-7.12		39118	40096	2.50	2	27157	28987	6.74		905	966		996	1041
6.00	6267	5715	-8.81		40158	41405	3.10	2	27624	29974	8.51		921	999		1013	1059
7.00	6409	5365	-16.28		42350	43548	2.83	2	29532	32817	11.13		984	1094		1083	1132
8.00	6535	4796	-26.61		43537	44604	2.45	2	30468	35012	14.92		1016	1167		1117	1168
9.00	6155	4067	-33.92		44387	45753	3.08	2	32078	37620	17.28		1069	1254		1176	1230
10.00	5380	3162	-41.23		46442	47575	2.44	2	35682	41251	15.61		1189	1375		1308	1368
11.00	5073	3475	-31.50		48589	49660	2.21	2	38443	42710	11.10		1281	1424		1410	1474
12.00	5164	3913	-24.24		50726	51917	2.35	2	40397	44092	9.15		1347	1470		1481	1549
13.00	4953	4091	-17.39		52806	54159	2.56	2	42901	45976	7.17		1430	1533		1573	1645
14.00	4786	3707	-22.55		54888	56355	2.67	2	45316	48941	8.00		1511	1631		1662	1737
15.00	5036	4426	-12.12		56620	58270	2.91	2	46549	49418	6.17		1552	1647		1707	1784
16.00	5736	5522	-3.72		57932	59565	2.82	2	46461	48521	4.43		1549	1617		1704	1781
17.00	6933	6367	-8.17		58640	59656	1.73	2	44773	46922	4.80		1492	1564		1642	1716
18.00	8355	6565	-21.42		57947	58574	1.08	2	41237	45443	10.20		1375	1515		1512	1581
19.00	9466	6764	-28.54		55627	55552	-0.14	2	36696	42024	14.52		1223	1401		1346	1407
20.00	10549	7652	-27.45		54124	53872	-0.47	2	33027	38568	16.77		1101	1286		1211	1266
21.00	12265	9712	-20.82		52358	52145	-0.41	2	27827	32721	17.59		928	1091		1020	1067
22.00	13679	11353	-17.00		50460	50343	-0.23	2	23102	27637	19.63		770	921		847	886
23.00	14866	13190	-11.27		47995	47832	-0.34	2	18263	21452	17.46		609	715		670	700
24.00	15867	14634	-7.77		45153	45176	0.05	2	13420	15907	18.54		447	530		492	514
max	15867	14634	-3.72		58640	59656	3.10		46549	49418	19.63		1552	1647			
min	4786	3162	-41.23		39118	40083	-0.47		13420	15907	3.64		447	530			

Day 2	DAM-Wind	RTM-Wind	% variation		DAM-Load	RTM-Load	% variation	SF	NL DAM	NL RAM	% variation		NL DAM (scaled)	NL RAM(scaled)		10% Reserve	15%Rsrv
1.00	16478	15147	-8.08		42682	42583	-0.23	2	9725	12289	26.37		972	950		1070	1118
2.00	16228	14499	-10.66		40335	40531	0.49	2	7879	11534	46.39		788	1153		867	906
3.00	15557	14135	-9.14		38622	39442	2.12	2	7507	11172	48.82		751	1117		826	863
4.00	14936	13924	-6.77		37564	38466	2.40	2	7693	10618	38.02		769	1062		846	885
5.00	14000	13187	-5.81		36982	38264	3.47	2	8982	11890	32.38		898	1189		988	1033
6.00	13058	12368	-5.29		37064	38295	3.32	2	10948	13559	23.85		1095	1356		1204	1259
7.00	12089	11572	-4.27		37726	38902	3.12	2	13548	15758	16.31		1355	1576		1490	1558
8.00	10991	10640	-3.20		38790	39479	1.78	2	16808	18200	8.28		1681	1820		1849	1933
9.00	9822	9445	-3.84		40281	40409	0.32	2	20637	21519	4.28		2064	2152		2270	2373
10.00	9091	8775	-3.47		42274	41732	-1.28	2	24093	24182	0.37		2409	2418		2650	2771
11.00	8693	8072	-7.14		43653	42800	-1.95	2	26268	26656	1.48		2627	2666		2889	3021
12.00	8013	6843	-14.60		44434	43194	-2.79	2	28407	29507	3.87		2841	2951		3125	3267
13.00	7238	5745	-20.63		44891	43871	-2.27	2	30415	32381	6.46		3041	3238		3346	3498
14.00	6671	5217	-21.80		45272	44509	-1.68	2	31929	34075	6.72		3193	3408		3512	3672
15.00	6302	4426	-29.77		45690	45168	-1.14	2	33085	36315	9.76		3309	3631		3639	3805
16.00	6141	4182	-31.89		46332	45963	-0.80	2	34051	37598	10.42		3405	3760		3746	3916
17.00	5963	4249	-28.75		46824	46640	-0.39	2	34898	38143	9.30		3490	3814		3839	4013
18.00	5769	3681	-36.19		46671	46692	0.04	2	35133	39329	11.94		3513	3933		3865	4040
19.00	5537	3420	-38.24		45519	45899	0.83	2	34446	39060	13.39		3445	3906		3789	3961
20.00	5542	4259	-23.14		45352	45603	0.55	2	34268	37085	8.22		3427	3708		3769	3941
21.00	6562	5426	-17.31		44536	44801	0.59	2	31413	33949	8.07		3141	3395		3455	3612
22.00	7498	5968	-20.40		43283	43413	0.30	2	28288	31477	11.28		2829	3148		3112	3253
23.00	8212	6849	-16.59		41601	42019	1.00	2	25177	28320	12.48		2518	2832		2769	2895
24.00	8804	7613	-13.53		39765	40133	0.93	2	22156	24907	12.42		2216	2491		2437	2548
max	16478	15147	-3.20		46824	46692	3.47		35133	39329	48.82		3513	3933			
min	5537	3420	-38.24		36982	38264	-2.79		7507	10618	0.37		751	950			



Ames implementation:

The simulation was performed using AMES software.

AMES was configured to perform:

- Day-ahead scheduling
- Real-time dispatch
- Reserve allocation
- Feasibility checking

The following reserve levels were analyzed:

- 0% reserve
- 10% reserve
- 15% reserve
- 50% reserve

Generator data, load data, and operational constraints were configured using the AMES input file.

Simulation procedure:

The simulation follows the following process:

1. Forecast load is used for Day-Ahead scheduling
2. Generator commitment is determined
3. Real-time load is introduced
4. System adjusts generation
5. Feasibility is checked
6. Results are analyzed

Results and discussion:

Case 1: 0% Reserve:

```
// FILENAME 2BusTestCase.dat
// UNIT SI
BASE_S 100
BASE_V 10

// Simulation Parameters
FNCSActive true
MaxDay 4

//Number of LSEs
NLSE 2

RTOPDur 60
RTKDur 60

BalPenPos 1000000
BalPenNeg 1000000

#ReservePercentStart
DARRD DARRU RTRRD RTRRU
0.0 0.0 0 0
#ReservePercentEnd

PriceSensitiveDemandFlag 0

// Bus Data
NumberOfBuses 2

#LineDataStart
// Name From To MaxCap Reactance
Branch1 1 2 1500 0.1000
#LineDataEnd

#GenDataStart
// Name ID atBus SCost a b c NS PMin PMax InitMoney
GenCo1 1 1 30.0000 100 15.0000 0.000 1 0.0000 5000.000 1000000.0000
GenCo2 2 2 10.0000 100 5.0000 0.000 1 0.0000 1000.000 1000000.0000
#GenDataEnd

#GenCostStart
// Name ColdStartTime ColdStartUpCost HotStartUpCost ShutDownCost
GenCo1 3 1500 0 0
GenCo2 3 2500 0 0
#GenCostEnd

#ScucInputDataStart
// Name PowerT0 UnitOnT0State MinUpTime MinDownTime NominalRampUp NominalRampDown StartupRampLim ShutdownRampLim
GenCo1 0.0000 -1 0 0 5000 5000 5000 5000
GenCo2 0.0000 -4 0 0 5000 5000 5000 5000
#ScucInputDataEnd

#LSEDataFixedDemandStart
// Name ID atBus H-00 H-01 H-02 H-03 H-04 H-05 H-06 H-07
LSE1 1 1 705.00 684.00 684.00 687.00 697.00 711.00 760.00 779.00
// Name ID atBus H-08 H-09 H-10 H-11 H-12 H-13 H-14 H-15
LSE1 1 1 797.00 849.00 903.00 961.00 1021.00 1064.00 1084.00 1091.00
// Name ID atBus H-16 H-17 H-18 H-19 H-20 H-21 H-22 H-23
LSE1 1 1 1086.00 1057.00 1030.00 1053.00 1043.00 1030.00 977.00 907.00
#LSEDataFixedDemandEnd
```

Day: 2 Hour: 6 Min: 0 Interval: 5 Count: 2

time requested 1 : 190802 time granted 1: 190802 Final time granted 1 : 190802

Waiting to receive bids/offers for RTM operation at count: 2

Day: 2 Hour: 6 Min: 0 Interval: 5 Count: 3

time requested 1 : 190803 time granted 1: 190803 Final time granted 1 : 190803

Waiting to receive bids/offers for RTM operation at count: 3

Day: 2 Hour: 6 Min: 0 Interval: 5 Count: 4

time requested 1 : 190804 time granted 1: 190804 Final time granted 1 : 190804

Waiting to receive bids/offers for RTM operation at count: 4

Day: 2 Hour: 6 Min: 0 Interval: 5 Count: 5

time requested 1 : 190805 time granted 1: 190805 Final time granted 1 : 190805

Waiting to receive bids/offers for RTM operation at count: 5

Bids/offers receiving window is closed.

RTM Optimization executing on Day 2 at Hour 6- Interval 6

Here is the standard output of the command with SCED:

SCED: Running RTM SCED using Modified version of PSST

SCED: Solver : cplex

SCED: Solver script file=D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmpfi7kreir.cplex.script

SCED: Solver log file: 'D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmptt3qiocn.cplex.log'

SCED: Solver solution file: 'D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmpv5h7enw9.cplex.sol'

SCED: Solver problem files: ('D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmp3lejg15n.pyomo.lp',)

SCED: TC :
SCED: infeasible
Solution is Infeasible because the reserve constraints are not satisfied.
Simulation is stopped.

The system became infeasible during Day 2 Hour 6 due to lack of reserve.

Case 2: 10% Reserve:

```
// FILENAME 2BusTestCase.dat
// UNIT SI
BASE_S 100
BASE_V 10

// Simulation Parameters
FNCSActive true
MaxDay 4

//Number of LSEs
NLSE 2

RTOPDur 60
RTKDur 60

BalPenPos 1000000
BalPenNeg 1000000

#ReservePercentStart
DARRD DARRU RTRRD RTRRU
0.1 0.1 0 0
#ReservePercentEnd

PriceSensitiveDemandFlag 0

// Bus Data
NumberOfBuses 2

#LineDataStart
// Name From To MaxCap Reactance
Branch1 1 2 1500 0.1000
#LineDataEnd

#GenDataStart
// Name ID atBus SCost a b c NS PMin PMax InitMoney
GenCo1 1 1 30.0000 100 15.0000 0.000 1 0.0000 5000.000 1000000.0000
GenCo2 2 2 10.0000 100 5.0000 0.000 1 0.0000 1000.000 1000000.0000
#GenDataEnd

#GenCostStart
// Name ColdStartTime ColdStartUpCost HotStartUpCost ShutDownCost
GenCo1 3 1500 0 0
GenCo2 3 2500 0 0
#GenCostEnd

#ScucInputDataStart
// Name PowerT0 UnitOnT0State MinUpTime MinDownTime NominalRampUp NominalRampDown StartupRampLim ShutdownRampLim
GenCo1 0.0000 -1 0 0 5000 5000 5000 5000
GenCo2 0.0000 -4 0 0 5000 5000 5000 5000
#ScucInputDataEnd

#LSEDataFixedDemandStart
// Name ID atBus H-00 H-01 H-02 H-03 H-04 H-05 H-06 H-07
LSE1 1 1 atBus 705.00 684.00 684.00 687.00 697.00 711.00 760.00 779.00
// Name ID atBus H-08 H-09 H-10 H-11 H-12 H-13 H-14 H-15
LSE1 1 1 atBus 797.00 849.00 903.00 961.00 1021.00 1064.00 1084.00 1091.00
// Name ID atBus H-16 H-17 H-18 H-19 H-20 H-21 H-22 H-23
LSE1 1 1 atBus 1086.00 1057.00 1030.00 1053.00 1043.00 1030.00 977.00 907.00
#LSEDataFixedDemandEnd
```

Day: 3 Hour: 1 Min: 0 Interval: 0 Count: 1
time requested 1 : 259201 time granted 1: 259201 Final time granted 1 : 259201
Waiting to receive bids/offers for RTM operation at count: 1
Received RTM loadforecast: 1153.0

Day: 3 Hour: 1 Min: 0 Interval: 0 Count: 2
time requested 1 : 259202 time granted 1: 259202 Final time granted 1 : 259202
Waiting to receive bids/offers for RTM operation at count: 2

Day: 3 Hour: 1 Min: 0 Interval: 0 Count: 3
time requested 1 : 259203 time granted 1: 259203 Final time granted 1 : 259203
Waiting to receive bids/offers for RTM operation at count: 3

Day: 3 Hour: 1 Min: 0 Interval: 0 Count: 4
time requested 1 : 259204 time granted 1: 259204 Final time granted 1 : 259204
Waiting to receive bids/offers for RTM operation at count: 4

Day: 3 Hour: 1 Min: 0 Interval: 0 Count: 5

time requested 1 : 259205 time granted 1: 259205 Final time granted 1 : 259205
Waiting to receive bids/offers for RTM operation at count: 5
Bids/offers receiving window is closed.

RTM Optimization executing on Day 3 at Hour 1- Interval 1

Here is the standard output of the command with SCED:

SCED: Running RTM SCED using Modified version of PSST
SCED: Solver : cplex
SCED: Solver script file=D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmp6d9_1jen.cplex.script
SCED: Solver log file: 'D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmp45vvjv37.cplex.log'
SCED: Solver solution file: 'D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmpxk9m5joz.cplex.sol'
SCED: Solver problem files: ('D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmph5h68n7_.pyomo.lp',)
SCED: TC :
SCED: infeasible
Solution is Infeasible because the reserve constraints are not satisfied.
Simulation is stopped.
JNIFncs end is called

The system operated successfully for longer periods but failed (infeasible) during high uncertainty conditions in Day 3.

Case 3: 15% Reserve:

```
// FILENAME 2BusTestCase.dat

// UNIT SI
BASE_S 100
BASE_V 10

// Simulation Parameters
FNCSActive true
MaxDay 4

//Number of LSEs
NLSE 2

RTOPDur 60
RTKDUR 60

BalPenPos 1000000
BalPenNeg 1000000

#ReservePercentStart
DARRD DARRU RTRRD RTRRU
0.15 0.15 0 0
#ReservePercentEnd

PriceSensitiveDemandFlag 0

// Bus Data
NumberOfBuses 2

#LineDataStart
// Name From To MaxCap Reactance
Branch1 1 2 1500 0.1000
#LineDataEnd

#GenDataStart
// Name ID atBus SCost a b c NS PMin PMax InitMoney
GenCo1 1 1 30.0000 100 15.0000 0.000 1 0.0000 5000.000 1000000.0000
GenCo2 2 2 10.0000 100 5.0000 0.000 1 0.0000 1000.000 1000000.0000
#GenDataEnd

#GenCostStart
// Name ColdStartTime ColdStartUpCost HotStartUpCost ShutDownCost
GenCo1 3 1500 0 0
GenCo2 3 2500 0 0
#GenCostEnd

#ScucInputDataStart
// Name PowerT0 UnitOnT0State MinUpTime MinDownTime NominalRampUp NominalRampDown StartupRampLim ShutdownRampLim
GenCo1 0.0000 -1 0 0 5000 5000 5000 5000
GenCo2 0.0000 -4 0 0 5000 5000 5000 5000
#ScucInputDataEnd

#LSEDataFixedDemandStart
// Name ID atBus H-00 H-01 H-02 H-03 H-04 H-05 H-06 H-07
LSE1 1 1 705.00 684.00 684.00 687.00 697.00 711.00 760.00 779.00
// Name ID atBus H-08 H-09 H-10 H-11 H-12 H-13 H-14 H-15
LSE1 1 1 797.00 849.00 903.00 961.00 1021.00 1064.00 1084.00 1091.00
// Name ID atBus H-16 H-17 H-18 H-19 H-20 H-21 H-22 H-23
LSE1 1 1 1086.00 1057.00 1030.00 1053.00 1043.00 1030.00 977.00 907.00
#LSEDataFixedDemandEnd
```

Day: 3 Hour: 2 Min: 0 Interval: 1 Count: 1

time requested 1 : 262801 time granted 1: 262801 Final time granted 1 : 262801

Waiting to receive bids/offers for RTM operation at count: 1
Received RTM loadforecast: 1117.0

Day: 3 Hour: 2 Min: 0 Interval: 1 Count: 2
time requested 1 : 262802 time granted 1: 262802 Final time granted 1 : 262802
Waiting to receive bids/offers for RTM operation at count: 2

Day: 3 Hour: 2 Min: 0 Interval: 1 Count: 3
time requested 1 : 262803 time granted 1: 262803 Final time granted 1 : 262803
Waiting to receive bids/offers for RTM operation at count: 3

Day: 3 Hour: 2 Min: 0 Interval: 1 Count: 4
time requested 1 : 262804 time granted 1: 262804 Final time granted 1 : 262804
Waiting to receive bids/offers for RTM operation at count: 4

Day: 3 Hour: 2 Min: 0 Interval: 1 Count: 5
time requested 1 : 262805 time granted 1: 262805 Final time granted 1 : 262805
Waiting to receive bids/offers for RTM operation at count: 5
Bids/offers receiving window is closed.

RTM Optimization executing on Day 3 at Hour 2- Interval 2

Here is the standard output of the command with SCED:

SCED: Running RTM SCED using Modified version of PSST
SCED: Solver : cplex
SCED: Solver script file=D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmpl6zjb_ew.cplex.script
SCED: Solver log file: 'D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmpm0ib93vd.cplex.log'
SCED: Solver solution file: 'D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmpvo5lop5s.cplex.sol'
SCED: Solver problem files: ('D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmpingzsob_.pyomo.lp',)
SCED: TC :
SCED: infeasible
Solution is Infeasible because the reserve constraints are not satisfied.
Simulation is stopped.
JNIFncs end is called

The system showed slightly improved performance compared to 10% reserve but still became infeasible during high variation periods.

Case 4: 50% Reserve:

```
// FILENAME 2BusTestCase.dat

// UNIT SI
BASE_S 100
BASE_V 10

// Simulation Parameters
FNCSActive true
MaxDay 4

//Number of LSEs
NLSE 2

RTOPDur 60
RTKDur 60

BalPenPos 1000000
BalPenNeg 1000000

#ReservePercentStart
DARRD DARRU RTRRD RTRRU
0.5 0.5 0 0
#ReservePercentEnd

PriceSensitiveDemandFlag 0

// Bus Data
NumberOfBuses 2

#LineDataStart
// Name From To MaxCap Reactance
Branch1 1 2 1500 0.1000
#LineDataEnd
```

```

#GenDataStart
// Name ID atBus SCost a b c NS PMin PMax InitMoney
GenCo1 1 1 30.0000 100 15.0000 0.000 1 0.0000 5000.000 1000000.0000
GenCo2 2 2 10.0000 100 5.0000 0.000 1 0.0000 1000.000 1000000.0000
#GenDataEnd

#GenCostStart
// Name ColdStartTime ColdStartUpCost HotStartUpCost ShutDownCost
GenCo1 3 1500 0 0
GenCo2 3 2500 0 0
#GenCostEnd

#ScucInputDataStart
// Name PowerT0 UnitOnT0State MinUpTime MinDownTime NominalRampUp NominalRampDown StartupRampLim ShutdownRampLim
GenCo1 0.0000 -1 0 0 5000 5000 5000 5000
GenCo2 0.0000 -4 0 0 5000 5000 5000 5000
#ScucInputDataEnd

#LSEDataFixedDemandStart
// Name ID atBus H-00 H-01 H-02 H-03 H-04 H-05 H-06 H-07
LSE1 1 1 705.00 684.00 684.00 687.00 697.00 711.00 760.00 779.00
// Name ID atBus H-08 H-09 H-10 H-11 H-12 H-13 H-14 H-15
LSE1 1 1 797.00 849.00 903.00 961.00 1021.00 1064.00 1084.00 1091.00
// Name ID atBus H-16 H-17 H-18 H-19 H-20 H-21 H-22 H-23
LSE1 1 1 1086.00 1057.00 1030.00 1053.00 1043.00 1030.00 977.00 907.00
#LSEDataFixedDemandEnd

```

Day: 3 Hour: 24 Min: 0 Interval: 23 Count: 1
time requested 1 : 342001 time granted 1: 342001 Final time granted 1 : 342001
Waiting to receive bids/offers for RTM operation at count: 1
Received RTM loadforecast: 853.0

Day: 3 Hour: 24 Min: 0 Interval: 23 Count: 2
time requested 1 : 342002 time granted 1: 342002 Final time granted 1 : 342002
Waiting to receive bids/offers for RTM operation at count: 2

Day: 3 Hour: 24 Min: 0 Interval: 23 Count: 3
time requested 1 : 342003 time granted 1: 342003 Final time granted 1 : 342003
Waiting to receive bids/offers for RTM operation at count: 3

Day: 3 Hour: 24 Min: 0 Interval: 23 Count: 4
time requested 1 : 342004 time granted 1: 342004 Final time granted 1 : 342004
Waiting to receive bids/offers for RTM operation at count: 4

Day: 3 Hour: 24 Min: 0 Interval: 23 Count: 5
time requested 1 : 342005 time granted 1: 342005 Final time granted 1 : 342005
Waiting to receive bids/offers for RTM operation at count: 5
Bids/offers receiving window is closed.

RTM Optimization executing on Day 3 at Hour 24- Interval 24

Here is the standard output of the command with SCED:

```

SCED: Running RTM SCED using Modified version of PSST
SCED: Solver : cplex
SCED: Solver script file=D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmp1jte792r.cplex.script
SCED: Solver log file: 'D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmppz57b7ke.cplex.log'
SCED: Solver solution file: 'D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmp9i39_484.cplex.sol'
SCED: Solver problem files: ('D:\AMES-V5.1\tempFiles\PyomoTempFiles\tmp6nwl_pd9.pyomo.lp',)
SCED: TC :
SCED: optimal
time requested 2 : 342060 Final time granted 2 : 342060

```

RTLMP LMP for Bus 1: 5
RTLMP LMP for Bus 2: 5

time requested 3 : 345600 Final time granted 3 : 345600

Simulation End time: Sat May 02 21:47:01 IST 2026

The current simulation run is concluded for 4 days

Size of the double array: 24

The system remained feasible throughout all operating periods. However, such a large reserve level is economically impractical.

This analysis shows that increasing reserve improves feasibility but fixed reserve levels fail under extreme uncertainty.

Infeasibility occurs when real-time deviation exceeds available reserve capacity.

$$(\Delta P > R)$$

Where:

- ΔP = Real-time deviation
- R = Available reserve

This highlights the limitations of fixed reserve allocation.

Conclusion:

This work analyzed the impact of uncertainty on economic dispatch under time-sequential operation. Results show that increasing uncertainty leads to infeasibility when reserve is insufficient. Fixed reserve levels such as 10% and 15% improve system performance but fail during high variation periods. Although 50% reserve ensures feasibility, it is not practical. The study demonstrates the importance of adaptive reserve planning for future power systems.

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